ALPACA AI TRADING AGENTS HACKATHON 2026 · OPTIONS ALPHA AGENTS

An options agent that refuses to trade

Sells volatility on SPY only when it is measurably expensive. Every decision it makes, including every refusal, is a signed artifact you can verify yourself.

Nilay Toshniwal · Solo entry

ITS FIRST LIVE RUN

Implied vol: **12.81** Trailing realised: **13.28**

Volatility was cheap. So it refused.

Most trading agents are built to trade. This one measures whether the premium is actually worth selling, and says no when it isn't.

This is not a bug. It is the product.

THE THESIS

Options are priced richer than the move that actually follows

We call that gap the volatility risk premium (VRP). It is a well-documented structural feature of index options: buyers of protection pay more than the realised outcome justifies, on average.

The agent sells that gap with a defined-risk iron condor on SPY, and only enters when the premium is measurably rich enough, at the exact strikes it is about to sell, to be worth the risk.

DEFINED-RISK CONDOR

Only sold **when VRP clears a pre-registered threshold**, refusal is the default, not the exception.

RESEARCH DISCIPLINE

Pre-registered before a single line of backtest code existed

1

Hypotheses fixed first

Six trials, one bar, committed to git before any result existed, provable from the commit timestamps.

2

Holdout sealed in code

2023–2026 data cannot be loaded without an explicit, logged unseal call. One shot, only after a trial clears its development bar.

3

Multiple-testing corrected

The pass bar accounts for every trial run, not just the one that happened to win.

WHY THIS EXISTS

Two prior projects by the same author died of the same disease: too many trials against too short a window.

"Few trials, long window" is the fix, not a lower bar.

HYPOTHESIS 1

The edge is real, and it clears the bar decisively

+3.68

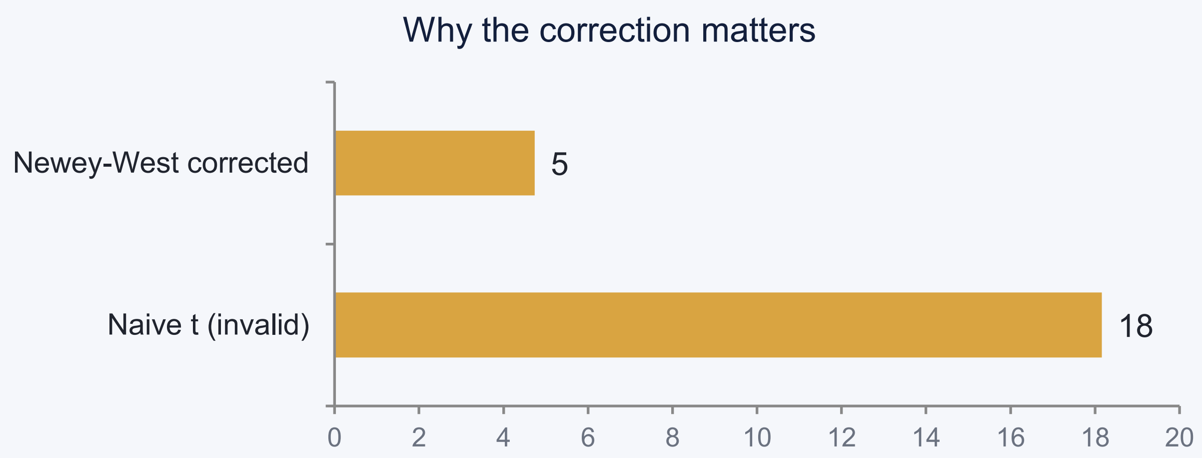
mean volatility risk premium, in annualised vol points

1,741

daily observations, 2016–2022 development window

t = +4.74

Newey-West corrected significance, one-sided $p < 0.0001$



21-day forward windows sampled daily overlap by 20 of 21 days, so the naive t-stat is invalid. The correction shrinks it ~4x. H1 still clears the bar with room to spare.

REPORTED WITH EQUAL WEIGHT

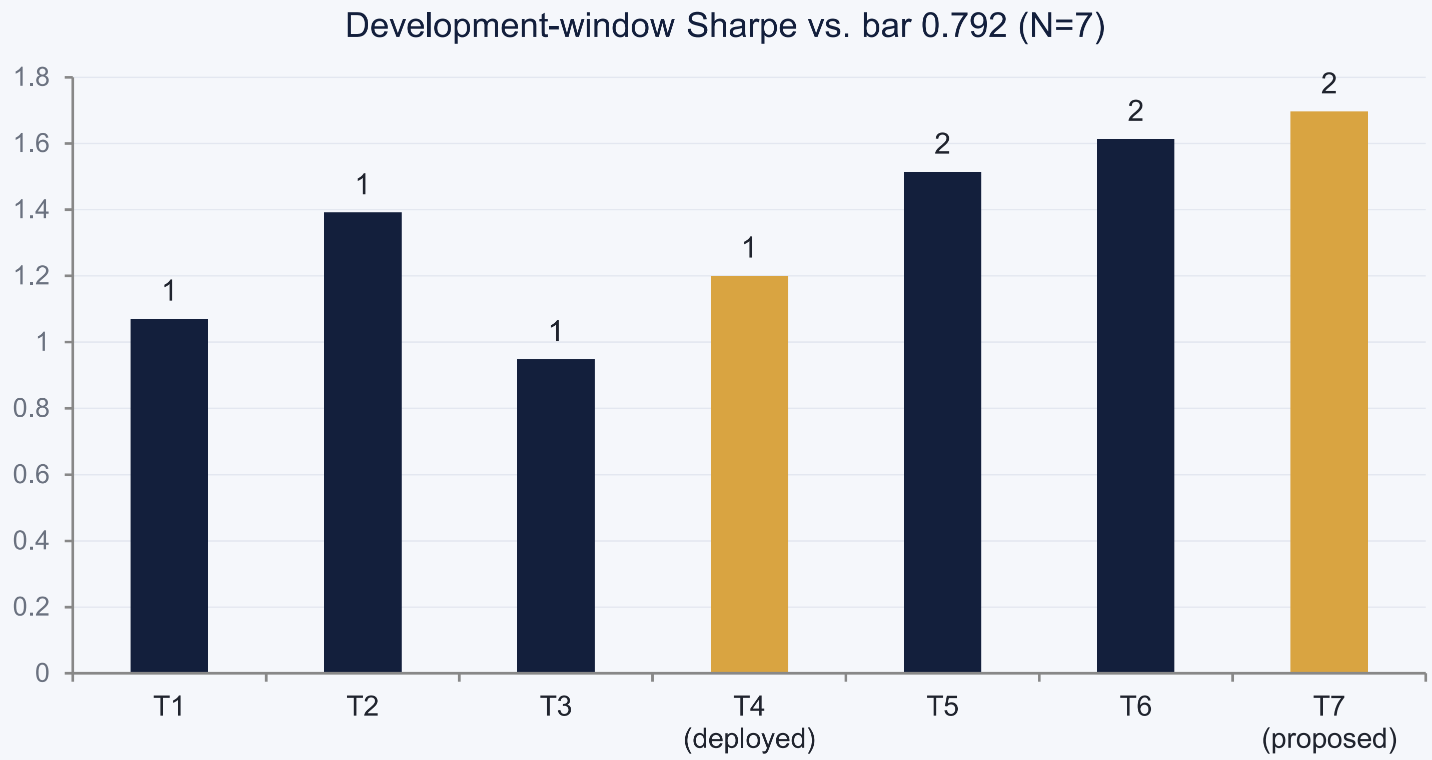
What the research found that did NOT work

Regime gate prediction	Predicted contango pays ≥1.0 pt more than backwardation. Measured: only +0.59. Recorded as WRONG, not dropped.
Pricing model, first pass	Failed its own pre-registered 15% error gate at 26.78%. VIX is a variance-swap rate, not ATM vol. Fixed and re-gated at 10.75%.
Best backtested trial	Doubling transaction costs improved its Sharpe (+1.614 to +1.728), impossible if the edge were clean. Disclosed as a fragility flag, not hidden.

A submission that only reports its wins is a sales document.

HYPOTHESIS 3 · SEVEN TRIALS, ONE BAR

The deployed tenor isn't the best-backtested one, on purpose



Both gold bars clear the bar. T7 (5–10 DTE) is the newest, best result and does not show T6's cost-anomaly. But T4 (7–14 DTE) is what's live, chosen for the 4.5-day judging window, not the backtest leaderboard:

A 42-day position captures only 7–9% of its credit in a 4.5-day hold. Tenor was chosen to fit the window the judges actually measure, not to maximise a backtest number.

ARCHITECTURE

Market data in, a decision out: always logged, win or refuse

Market data (MCP) → Signals (VRP, term structure) → 10 risk gates → Execute or refuse → Merkle artifact

Circuit breakers can halt the whole pipeline at any gate. A halt is not the same decision as a refusal, and both are recorded differently.

Real MCP, not REST

JSON-RPC over stdio, verified against Alpaca MCP Server v3.4.7, 74 tools. Every request/response is logged so the claim is checkable, not asserted.

VIX forward via put-call parity

The feed carries no index greeks, so the hedge recovers VIX's own forward price from real quotes before pricing the tail-risk call.

OS-level watchdog

Each cycle runs in its own subprocess with a hard OS timeout: an in-process timer can't rescue a thread stuck inside a syscall, but killing the process can.

Domain-separated Merkle tree

Leaves and nodes hash with distinct prefixes (0x00 / 0x01), closing the classic second-preimage attack a naive tree leaves open.

DEFENSE IN DEPTH

Five correlated positions need more than one line of defence

1

Circuit breakers

Daily loss limit, portfolio drawdown halt, consecutive-loss pause, distinguished in code from an ordinary refusal.

2

Tail hedge

Long VIX calls, 1% of equity, following Cboe’s VXTH methodology. Recovered via put-call parity since the feed carries no index greeks.

3

Sized on purpose

5 concurrent @ 3% (15% cap), a deliberate, recorded deviation from 1% research sizing. Hard-capped, never open-ended.

4

Staggered expiries

No two positions share an expiry. Five positions on one date is one position at 15% wearing a costume.

ORIGINAL ENGINEERING

Three strategies race live, in parallel, without risking a cent

Rather than commit to one tenor and hope, three configurations run as fully isolated shadow agents, hourly, against the real live market, through the agent's own valid trading window. Each is a genuine decision, just never sent as an order.

T4	T6	T7
7–14 DTE deployed	21–45 DTE old backtest-best	5–10 DTE new proposed best
↓	↓	↓
own isolated artifact log	own isolated artifact log	own isolated artifact log

↓ converge nightly ↓

Bonus, not the goal: building this comparison harness found and fixed 4 real production bugs before the live week could hit them: a Merkle-root path collision, a UTC/ET timestamp mismatch, and two separate silent scheduling failures.

TRUST, BUT VERIFY

Every decision is a signed artifact

```
VERIFIED: 14 artifacts, root 1a338b5a09ea90bd...
matches the seal written 2026-08-27T07:18:45Z
```

A SHA-256 Merkle tree over every decision: domain-separated leaves and nodes, root sealed before outcomes are known. Anyone can recompute it and confirm nothing was edited after the fact.

The judge doesn't have to trust the operator, which is the whole point when the operator has an incentive to look good.

VERIFY IT YOURSELF

github.com/nilaymastaadmi/alpaca-hackathon

Live dashboard → Streamlit Cloud

```
make test   →  195 tests
make verify → recomputes the root
```

THE UNCOMFORTABLE SLIDE

Will it even trade this week?

0 / 14

cycles logged so far have entered.
Every single one refused.

~28–33%

estimated chance of even ONE trade across the 4.5-day judged window, at the measured VRP distribution, and that's the optimistic side.

This is not a bug. The pre-registered research always expected roughly one trade per week at this sizing. Refusal is the strategy working, not failing. But P&L Performance is a judged criterion, and an agent that refuses all week scores zero on it.

We are not hiding this number, and we won't lower the threshold after a quiet Monday to manufacture a trade. If it changes, the reasoning gets written down BEFORE the number does.

31 AUG – 4 SEP

Live week results

DRAFT: UPDATE DURING THE LIVE WEEK

1. Dashboard screenshot: decisions, gates, artifact count
2. Trades entered vs. refused, with the VRP reading that decided each
3. Realised P&L, and whether the NFP de-risk (Thu 3 Sep close) fired
4. Final Merkle root and “make verify” output at submission time

WHAT MAKES THIS DIFFERENT

- Refusal is the product, not a fallback. Measured every cycle, logged every time
- Pre-registered before results existed, and the misses are reported as loudly as the wins
- A verifiable, tamper-evident decision trail: not a claim, a recomputation

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Built solo. Every claim measured, not assumed.

`github.com/nilaymastaadmi/alpaca-hackathon`

Live dashboard: Streamlit Cloud (link in submission)

Thank you.